

Generative Modeling with Variational Autoencoders

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Abstract. This report presents the design, implementation, and evaluation of a Variational Autoencoder (VAE) for generative modeling of the Fashion-MNIST dataset. The VAE is trained to learn a compressed latent representation of 28×28 grayscale clothing images and to generate new synthetic samples from that representation. The encoder maps each input image to a Gaussian distribution in latent space, parameterised by its mean μ and log-variance $\log \sigma^2$. The reparameterisation trick is used to draw differentiable samples, which the decoder maps back to pixel space. Training minimises a combined loss of Binary Cross-Entropy reconstruction error and KL divergence against a standard normal prior. The primary model uses a latent dimension of 20 and achieves a best validation loss of 240.72 after 30 epochs on a Tesla T4 GPU. An experimental study across five latent dimensions ($d \in \{2, 10, 20, 50, 100\}$) shows that $d = 20$ gives the best reconstruction quality (MSE = 13.29), while $d = 2$ degrades performance significantly (MSE = 22.18). Generated samples and t-SNE visualisations confirm that the model learns a structured and continuous latent space.

Keywords: Variational Autoencoder · Generative Modelling · Fashion-MNIST · Latent Space · Deep Learning

1 Introduction

Generative models learn the underlying distribution of a dataset so that new, plausible samples can be synthesised. Among deep generative models, the Variational Autoencoder (VAE) [1] provides a principled probabilistic framework: the encoder approximates the posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and the decoder models the likelihood $p_\theta(\mathbf{x}|\mathbf{z})$, while the KL divergence term acts as a regulariser to ensure the latent space remains smooth and suitable for sampling.

Fashion-MNIST [2] was chosen as the benchmark dataset because it is a direct drop-in replacement for the original MNIST handwritten digits, but is more challenging due to the higher within-class variation of clothing items. It consists of 70,000 grayscale 28×28 images spanning ten clothing categories.

This work addresses the following questions: (i) Can a fully-connected VAE learn to reconstruct and generate realistic Fashion-MNIST images? (ii) How does the choice of latent dimensionality affect reconstruction quality and the generative capability of the model?

The remainder of the report is organised as follows. Section 2 describes the dataset, architecture, loss function, and training procedure. Section 3 presents quantitative and qualitative results. Section 4 analyses the findings and discusses limitations. Section 5 summarises the work and outlines future directions.

2 Methodology

2.1 Dataset and Preprocessing

Fashion-MNIST contains 60,000 training images and 10,000 test images across ten balanced classes (T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot). Each image is a single-channel 28×28 pixel array stored as an 8-bit unsigned integer.

The following preprocessing steps were applied locally before training:

1. **Normalisation.** Pixel values were divided by 255 to map them into $[0, 1]$.
2. **Flattening.** Each 28×28 image was reshaped into a 784-dimensional vector, as the architecture uses fully connected layers.
3. **Dataset splitting.** The 60,000 training samples were split 90/10 into a training set of 54,000 images and a validation set of 6,000 images. The original 10,000-image test set was kept separate. A fixed random seed of 42 was used to ensure reproducibility.

Data was loaded in mini-batches of 128 samples using PyTorch `TensorDataset` and `DataLoader`, yielding 422 training batches, 47 validation batches, and 79 test batches per epoch.

2.2 VAE Architecture

The model follows the standard VAE architecture of Kingma and Welling [1], implemented as a fully connected network in PyTorch. The complete architecture for the primary model (latent dimension $d = 20$) is listed in Table 1 and contains 1,085,752 trainable parameters.

Encoder. The encoder $q_\phi(\mathbf{z}|\mathbf{x})$ maps an input vector $\mathbf{x} \in \mathbb{R}^{784}$ through two fully connected hidden layers (dimensions 512 and 256) with Batch Normalisation, ReLU activation, and Dropout ($p = 0.2$). Two separate linear layers then produce the latent mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and log-variance $\log \boldsymbol{\sigma}^2 \in \mathbb{R}^d$.

Reparameterisation trick. To allow gradients to flow through the stochastic sampling step, the latent vector is computed as

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (1)$$

where $\boldsymbol{\sigma} = \exp(0.5 \cdot \log \boldsymbol{\sigma}^2)$ and $\odot$ denotes element-wise multiplication. This formulation separates the deterministic parameters from the stochastic noise, making the operation differentiable with respect to $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$.

Table 1. VAE architecture for the primary model ($d = 20$). BN denotes Batch Normalisation; DO denotes Dropout with $p = 0.2$.

Component	Layer	Input dim	Output dim
4*Encoder	Linear + BN + ReLU + DO	784	512
	Linear + BN + ReLU + DO	512	256
	Linear (μ)	256	20
	Linear ($\log \sigma^2$)	256	20
Latent	Reparameterisation	20	20
4*Decoder	Linear + BN + ReLU + DO	20	256
	Linear + BN + ReLU + DO	256	512
	Linear + Sigmoid	512	784

Decoder. The decoder $p_\theta(\mathbf{x}|\mathbf{z})$ is the mirror of the encoder: it maps $\mathbf{z} \in \mathbb{R}^d$ back through two fully connected layers (256 then 512 units), each followed by Batch Normalisation, ReLU, and Dropout, and then to a 784-dimensional output via a linear layer with sigmoid activation, producing pixel values in $[0, 1]$.

Generation. New samples are generated by drawing $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and passing them through the decoder.

2.3 Loss Function

The VAE objective [1] is the Evidence Lower BOund (ELBO), which is maximised during training. Equivalently, we minimise the negative ELBO:

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \underbrace{\mathcal{L}_{\text{recon}}}_{\text{reconstruction}} + \beta \cdot \underbrace{\mathcal{L}_{\text{KL}}}_{\text{regularisation}}. \quad (2)$$

Reconstruction loss. Binary Cross-Entropy (BCE) is used, summed over all pixels:

$$\mathcal{L}_{\text{recon}} = - \sum_{i=1}^{784} \left[x_i \log \hat{x}_i + (1 - x_i) \log(1 - \hat{x}_i) \right]. \quad (3)$$

KL divergence. With Gaussian encoder and standard normal prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, the KL term has a closed form:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right). \quad (4)$$

Both terms are summed (not averaged) over pixels and latent dimensions, and then averaged over the batch. The weight $\beta = 1.0$ corresponds to the standard VAE formulation. All loss values reported in this paper are averaged over the number of samples in the respective split.

Table 2. Hyperparameters for the primary model.

Hyperparameter	Value
Optimiser	Adam
Learning rate	1×10^{-3}
Batch size	128
Epochs	30
Latent dim (d)	20
Hidden dims	[512, 256]
β (KL weight)	1.0
Dropout	0.2
Random seed	42

Table 3. Training and validation losses at selected epochs for the primary model ($d = 20, 30$ epochs). Losses are per-sample averages.

Epoch	Train			Validation		
	Total	Recon	KL	Total	Recon	KL
1	280.68	268.99	11.69	257.01	245.62	11.38
5	253.05	241.00	12.05	246.21	234.30	11.91
10	250.02	237.57	12.45	243.30	231.06	12.24
15	248.72	236.12	12.60	242.11	229.66	12.44
20	247.90	235.21	12.69	241.47	229.06	12.41
25	247.32	234.56	12.75	241.01	228.56	12.46
30	247.00	234.22	12.78	240.90	228.56	12.34
Best validation loss				240.72		

2.4 Training Procedure

Training was performed on a Google Colab instance equipped with a Tesla T4 GPU (15.64 GB VRAM). The full set of hyperparameters is listed in Table 2.

The model with the lowest validation loss across the 30 epochs was saved as the best checkpoint. Total training time was approximately 1.62 minutes. For the experimental study (Section 3.5), each of the five models was trained for 20 epochs, with all other hyperparameters held constant.

3 Experimental Results

3.1 Training Dynamics

Table 3 reports the training and validation losses at selected epochs for the primary model ($d = 20$). Both the total loss and its components are shown.

The reconstruction loss decreases steadily and accounts for the majority of the total loss at every epoch. The KL divergence starts low (11.38 at epoch 1) and rises gradually to approximately 12.4–12.8, which is the expected behaviour:

the model first learns to reconstruct well and then progressively regularises the latent space. The gap between training and validation loss is small throughout, indicating that the model generalises rather than overfitting.

3.2 Image Reconstruction

Qualitative reconstruction results are shown in Fig. 1. The model reconstructs the overall shape and texture of each garment correctly. Fine details such as individual stitching or logo text are not reproduced, which is typical for pixel-wise BCE-trained VAEs because the loss averages over all possible reconstructions consistent with the data.

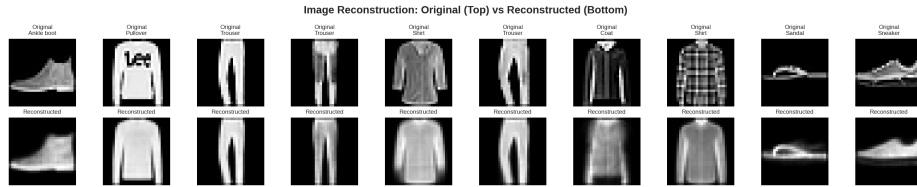


Fig. 1. Top row: original test images. Bottom row: reconstructions produced by the primary model ($d = 20$).

3.3 Image Generation

Figure 2 shows 20 images generated by sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and passing each sample through the decoder. The generated images display recognisable garment silhouettes, including items that resemble trousers, boots, bags, and tops. The diversity of the generated samples suggests the model has not collapsed to a single mode.

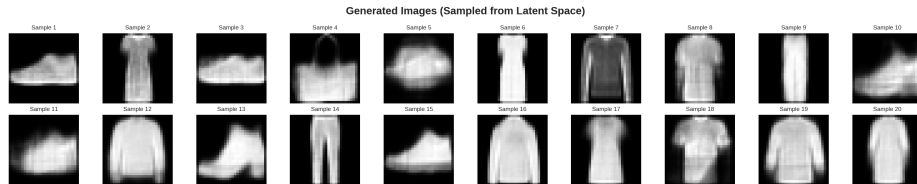


Fig. 2. Twenty images generated by sampling from the standard normal prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and decoding.

3.4 Latent Space Structure

Figure 3 shows a t-SNE projection of the encoder means μ for 2,000 test images, coloured by class label. Despite the high overlap expected from a standard VAE (which trades disentanglement for coverage), adjacent clothing categories form loose but distinguishable clusters. Footwear classes (Sandal, Sneaker, Ankle boot) are grouped together, as are upper-body clothing items (Pullover, Coat, Shirt). T-shirt and Shirt show the most overlap, which is consistent with their visual similarity.



Fig. 3. t-SNE visualisation of latent means μ for 2,000 test images, coloured by clothing category.

3.5 Experimental Study: Impact of Latent Dimension

Five VAE models were trained with identical hyperparameters but varying latent dimensions $d \in \{2, 10, 20, 50, 100\}$. Each model was trained for 20 epochs. Table 4 summarises the results; MSE and PSNR are computed on the test set.

Key observations. $d = 2$ is clearly insufficient. With only two latent dimensions, the model must compress all variation of ten clothing categories into a two-dimensional space. Validation loss is 260.97, compared to 241.56 for $d = 20$

Table 4. Comparison of VAE models across five latent dimensions. All models share the same hidden dimensions [512, 256], learning rate 10^{-3} , and 20 training epochs. MSE and PSNR (computed as $10 \log_{10}(1/\text{MSE})$) are per-pixel averages on the test set.

d	Parameters	Train Loss	Val Loss	Best Val Loss	Time (s)	MSE	PSNR (dB)
2	1,071,892	266.64	260.97	260.97	61.37	22.183	−13.46
10	1,078,052	247.86	241.64	241.64	62.40	13.355	−11.26
20	1,085,752	247.99	241.56	241.56	62.52	13.292	−11.24
50	1,108,852	248.13	241.85	241.80	61.42	13.445	−11.29
100	1,147,352	248.81	242.80	242.65	63.46	14.035	−11.47

— a gap of 19.4 loss units. MSE is 22.18, almost $1.7\times$ higher than the best model.

$d = 20$ **achieves the best reconstruction quality.** It produces the lowest MSE (13.29) and PSNR (−11.24 dB). In terms of validation loss, $d = 20$ and $d = 10$ are very close (241.56 vs. 241.64), but $d = 20$ has a slight edge.

Performance degrades beyond $d = 20$. Both $d = 50$ and $d = 100$ produce slightly higher MSE and lower PSNR than $d = 20$. This is likely because a higher-dimensional latent space makes it harder for the KL regulariser to enforce a compact, well-shaped prior, so sampled vectors at test time fall into regions of the latent space that the decoder has not been trained on. This is consistent with the known difficulty of sampling from high-dimensional VAEs [1].

Training time is practically constant. All five models train in approximately 61–64 seconds for 20 epochs, showing that the latent dimension does not dominate computation time compared to the large hidden layers.

Figure 4 shows these trends graphically.

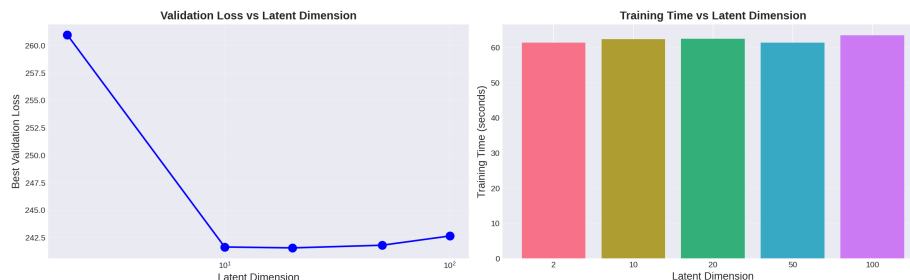


Fig. 4. Left: best validation loss as a function of latent dimension. Right: training time in seconds. Both axes use the latent dimensions $d \in \{2, 10, 20, 50, 100\}$.

4 Discussion

4.1 Interpretation of the Learned Latent Space

The t-SNE visualisation (Fig. 3) and interpolation together show that the VAE has learned a structured latent space. Semantically similar classes are grouped near each other: footwear, bags, and upper-body clothing each form separate neighbourhoods. This structure is not explicitly supervised; it emerges because the encoder must use similar latent codes for images that require similar pixel patterns.

The smooth interpolation in Fig. ?? confirms that the latent space is continuous: decoding a straight-line path between two encoded points produces images that gradually change from one garment to another without discontinuities. This is a key property that distinguishes VAEs from standard autoencoders, where random points in latent space may not decode to meaningful images.

4.2 Limitations

Blurry reconstructions. The BCE loss penalises each pixel independently, so the model tends to produce the average over all plausible reconstructions consistent with the latent code. This causes the reconstructed images to appear smoother and less sharp than the originals. This is a well-known limitation of pixel-wise likelihoods in VAEs.

Fixed encoder architecture. The encoder and decoder use only fully connected layers. Spatial structure in images is encoded through the weights of the first linear layer, but no explicit convolutional inductive bias is used. A convolutional encoder-decoder would be better suited to preserve local spatial patterns in the images.

Gaussian prior assumption. The model assumes the prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, which is unimodal. For a dataset with ten distinct clothing categories, a mixture prior may be more appropriate and could improve both reconstruction and generation quality.

Fixed β . We used $\beta = 1$ throughout. Higher values of β would produce more disentangled latent representations [3] at the cost of reconstruction quality, while lower values would improve reconstruction but reduce structure in the latent space. This trade-off was not explored in this work.

No standard generative quality metric. We report MSE and PSNR as proxy metrics for reconstruction quality, but these do not measure perceptual quality or sample diversity. Metrics such as the Fréchet Inception Distance (FID) or Inception Score (IS) would provide a more complete evaluation of generative performance.

4.3 Possible Improvements

Convolutional architecture (Conv-VAE). Replacing the fully connected layers with convolutional and transposed convolutional layers would allow the model to exploit spatial structure in the images and should reduce blurriness.

KL annealing. Gradually increasing the weight of the KL term from zero to $\beta = 1$ over the first several epochs prevents the encoder from collapsing into the prior early in training and typically improves both reconstruction and latent structure.

Conditional VAE (CVAE). Conditioning both the encoder and decoder on the class label would allow class-specific generation and would reduce overlap between classes in latent space.

Adversarial training (VAE-GAN). Adding a discriminator to the training objective can sharpen the generated images by penalising blurry outputs at the feature level rather than at the pixel level.

Better evaluation. Computing FID on the generated images would provide a standardised metric for comparing the VAE against other generative models on Fashion-MNIST.

5 Conclusion

This report presented a fully connected VAE trained on Fashion-MNIST for generative modelling of clothing images. The model was implemented in PyTorch and trained with an Adam optimiser, BCE reconstruction loss, and a KL divergence regulariser. The primary model ($d = 20$, 1,085,752 parameters) achieved a best validation loss of 240.72 after 30 epochs and produced visually faithful reconstructions and diverse generated samples within 1.62 minutes of training on a T4 GPU.

An experimental study across five latent dimensions showed that $d = 20$ is the best setting for Fashion-MNIST under the given architecture: it achieves the lowest test MSE (13.29) and PSNR (-11.24 dB), while $d = 2$ produces significantly worse results and $d \geq 50$ shows marginal degradation. Latent space analysis via t-SNE and interpolation confirmed that the model learns a structured and continuous representation.

Future work should explore convolutional architectures to reduce blurriness, KL annealing to improve latent structure, conditional generation for class-specific control, and FID-based evaluation for more rigorous quantitative comparison.

Disclosure of Interests. The authors have no competing interests to declare.

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